

Academic Year: 2025 - 26

Module Code and Module Title: COMP1814 Statistical Techniques with R

Module Leader: Dr Konstantinos Skindilias

Due date: Monday 23 March 2026 by 17:00 UK Local Time

Student Workload:

This coursework is expected to take an average student who is up to date with tutorial work approximately 15 hours to complete.

Feedback Release: Feedback and grades for the main sit will be released on 19/05/2026. For the late sit, feedback and grades are normally released within 27 working days of the original deadline.

Learning Outcomes: ALL

Plagiarism and AI declaration

- **Guidance to the AI usage and the university policy and student declaration: [guidance to AI usage](#); [declaration of AI use](#).**

Coursework Submission Requirements

- **The last version you upload will be the one that is marked.**
- **For this coursework you must submit a single Acrobat PDF document. In general, any text in the document must not be an image (i.e. must not be scanned) and would normally be generated from other documents (e.g. MS Office using "Save As .. PDF").**
- **There are limits on the file size. The current limits are displayed on the Moodle coursework submission page.**
- **Make sure that any files you upload are virus-free and not protected by a password or corrupted otherwise they will be treated as null submissions.**

Year 2 Undergraduate Coursework: Hypothesis Testing in R

Deadline: Monday 23 March 2026 by 17:00 UK Local Time

Dataset: Run the provided R script (bottom of page) to generate your **unique dataset** before beginning the analysis.

Instructions for Students

1. **Page Limit:** Your report must not exceed **4 pages**, including figures and tables. Ensure conciseness while covering all necessary details.
 2. **Creativity in Presentation:** You are encouraged to be **creative** in how you present your answers. Use **tables, figures, and well-structured sections** to enhance readability.
 3. **Displaying Code:** Clearly separate code and explanations. Specifically:
 - Show only **essential code** responsible for generating **(i) plots and figures** and **(ii) statistical tests**.
 - Avoid unnecessary raw outputs or redundant code blocks.
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Coursework Questions and Marking Breakdown

Section 1: Exploring and Understanding the Data (15 marks)

1. **Load your unique dataset** into R and display its structure. (5 marks)
 - How many observations and variables are present?
 - What are the data types of each variable?
2. **Perform data cleaning and transformations** (10 marks)
 - Convert relevant categorical variables into factors.
 - Summarize the key statistics (mean, median, standard deviation) of numerical variables.

Section 2: Comparing Means Between Two Groups (20 marks)

3. **Create a boxplot** comparing **birth weight** (`birthwt.grams`) between **smoking and non-smoking mothers**. (5 marks)
 - What trends do you observe?
 - Are there any potential **outliers**?
4. **Compute the mean, standard deviation, and standard error** of birth weight for smoking vs. non-smoking mothers. (5 marks)
 - Summarize your findings in a table.
5. **Perform an independent two-sample t-test** to assess whether smoking significantly affects birth weight. (10 marks)
 - Report the **t-statistic**, **p-value**, and **confidence interval**.
 - **Interpret the results:** Is the difference statistically significant at a **5% level**?

Section 3: Checking Normality and Alternative Tests (15 marks)

6. **Assess normality** of birth weight using: (7 marks)
 - **QQ-plots:** Do the data points follow a straight line?

- **Histograms:** Do the distributions appear normal or skewed?
- 7. **If the normality assumption is violated, conduct a Wilcoxon rank-sum test (`wilcox.test()`).** (8 marks)
 - Compare the p-value from the Wilcoxon test to the t-test.
 - **Does it change your conclusion?**

Section 4: Hypothesis Testing for Categorical Data (20 marks)

- 8. **Create a 2×2 contingency table** showing the relationship between: (5 marks)
 - **Low birth weight** (`birthwt.below.2500`) and **smoking status**.
 - How many mothers in each group had low birth weight babies?
- 9. **Conduct Fisher's exact test and chi-squared test to examine the association.** (10 marks)
 - **Compare the p-values:** Do both tests yield similar conclusions?
- 10. **Interpret the results:** Does smoking significantly increase the risk of low birth weight? (5 marks)

Section 5: Simulation and Understanding P-values (20 marks)

- 11. **Simulate a dataset** with a control and treatment group where the treatment has **no effect**. (8 marks)
 - Run a t-test and report the p-value.
 - **What does this tell you about false positives?**
- 12. **Repeat the simulation 10,000 times, but now with a moderate treatment effect.** (12 marks)
 - Plot a histogram of the **p-values**.
 - **What does this tell us about statistical power and detecting real effects?**

Bonus Question (Optional for High Achievers) (10 marks)

- 13. **What if the sample size was very small?** (10 marks)
 - Re-run the t-test and Wilcoxon test using only **30 random observations**.
 - Compare results: **Does the choice of test matter more when sample sizes are small?**

Presentation & Code Quality (10 marks)

- **Report clarity, structure, and explanations** (5 marks)
- **Code readability, efficiency, and reproducibility** (5 marks)

TOTAL MARKS: 100 (110 with bonus) – capped at 100

Submission Guidelines

- Submit a **well-structured report** including:
 - **Code snippets** with explanations.
 - **Plots and tables** to support your findings.
 - **Interpretations of results** in clear, non-technical language.
 - **Submit your dataset (birthwt_student_[ID].csv) along with your report.**
 - Ensure that **all results are reproducible** by providing a clean R script.
 - **Classification is driven by depth of critical evaluation:** Reproducing textbook procedures corresponds to a 2:2, competent application with some explanation to a 2:1, strong independent analysis beyond textbook material to a 1st, and exceptional originality, synthesis, and critical insight to a high 1st.
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What This Coursework Assesses

- ✓ **Understanding of hypothesis testing** in R.
- ✓ **Ability to interpret statistical results** beyond just reporting p-values.
- ✓ **Correct application of statistical tests** based on data assumptions.
- ✓ **R programming skills**, including visualization, data wrangling, and simulations.

Database instructions for Students

1. **Set your student ID** in the script below.
2. Run the script in **RStudio** or any R environment.
3. It will generate a **randomized version** of the `birthwt` dataset for your coursework.
4. Use this dataset for all your analysis tasks.

```
# Load necessary libraries
if (!require(MASS)) install.packages("MASS", dependencies=TRUE)
if (!require(dplyr)) install.packages("dplyr", dependencies=TRUE)
if (!require(tibble)) install.packages("tibble", dependencies=TRUE)

library(MASS)    # For birthwt dataset
library(dplyr)   # For data manipulation
library(tibble)  # For tidy data handling

# Step 1: Set your unique Student ID (Change this!)
student_id <- 123456 # ! Replace with your own Student ID (numbers only) !

# Step 2: Load and clean the birthwt dataset
set.seed(student_id) # Ensure results are unique but reproducible
birthwt <- as_tibble(MASS::birthwt)

# Rename variables for clarity
birthwt <- birthwt %>%
  rename(
    birthwt.below.2500 = low,
    mother.age = age,
    mother.weight = lwt,
```

```

    mother.smokes = smoke,
    previous.prem.labor = ptl,
    hypertension = ht,
    uterine.irr = ui,
    physician.visits = ftv,
    birthwt.grams = bwt
  )

# Convert categorical variables to factors with meaningful labels
birthwt <- birthwt %>%
  mutate(
    race = recode_factor(race, `1` = "white", `2` = "black", `3` = "other"),
    mother.smokes = recode_factor(mother.smokes, `0` = "no", `1` = "yes"),
    hypertension = recode_factor(hypertension, `0` = "no", `1` = "yes"),
    uterine.irr = recode_factor(uterine.irr, `0` = "no", `1` = "yes"),
    birthwt.below.2500 = recode_factor(birthwt.below.2500, `0` = "no", `1` =
"yes")
  )

# 🚩 Step 3: Randomize the dataset to ensure unique student datasets

# (A) Randomly sample a subset of 100 rows for each student
birthwt_sample <- birthwt %>%
  sample_n(100, replace = FALSE)

# (B) Add slight random noise to birth weight values
birthwt_sample <- birthwt_sample %>%
  mutate(birthwt.grams = birthwt.grams + rnorm(n(), mean = 0, sd = 20))

# (C) Shuffle the smoking status to prevent direct comparison
birthwt_sample <- birthwt_sample %>%
  mutate(mother.smokes = sample(mother.smokes))

# (D) Randomize variable names for some numerical columns
random_labels <- sample(c("factorA", "factorB", "factorC"))
colnames(birthwt_sample)[c(3, 4, 5)] <- random_labels

# 🚩 Step 4: Save the unique dataset
write.csv(birthwt_sample, file = paste0("birthwt_student_", student_id, ".csv"),
row.names = FALSE)

# 🚩 Step 5: Display confirmation message
cat("\n✅ Your unique dataset has been saved as:", paste0("birthwt_student_",
student_id, ".csv"))
cat("\nUse this dataset for your coursework and analysis!\n")

# 🚩 Step 6: Show the first few rows
print(head(birthwt_sample))

```